

Diffraction

Laboratory Experiment #2

By

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Submitted To:

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Lab Report

1.) Objective

The objective and intent of our group in this experiment was to investigate the image formed by a single slit 100 μm width, the image formed by a single round aperture of 300 μm diameter, and the images formed by a diffraction grating.

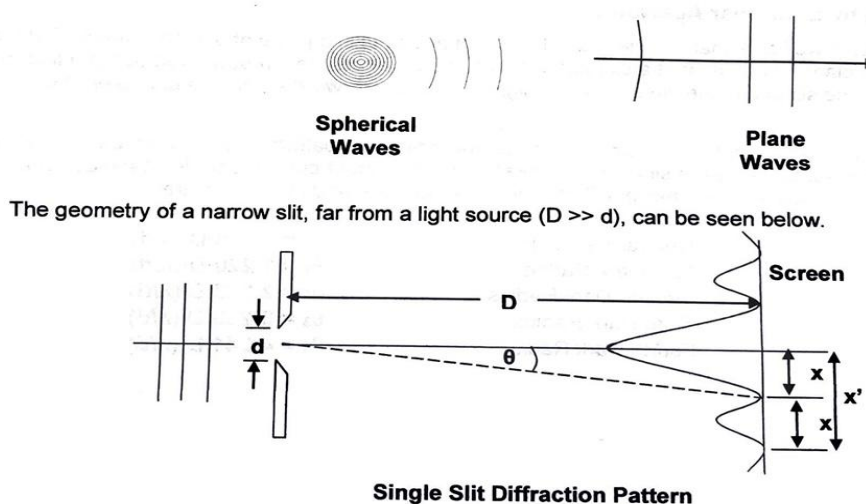
2.) Materials

- Optical Table (SG-22-2)
- Damped Rod
- TEM₀₀ HeNe Laser
- Adjustable Mounting Fixtures for HeNe Laser
 - Laser Mount (ULM-TILT)
 - Rod Clamp (340-RC)
- 2x Post Holders (VPH-2)
- Goggles
- 1" Dia. Adj. Mirror Mount (E55-532)
- 8.5x11 Paper
- 300 μm Pinhole (NT39-544)
- Masking Tape
- 100 μm Air Slit (NT58-544)
- 300 μm Pinhole
- Meter Stick and Plastic Calipers
- Screw Kit, $\frac{1}{4}$ x 20 (SK-25A)

3.) Background

Background- Single Slit Diffraction

Since a laser beam, such as from a HeNe laser, has such a small divergence we may consider the output wave motion to be almost like a plane wave. A plane wave is a spherical or "pond" wave very far away from its source much like a pond ripple very far from where the stone was dropped. A plane wave can be represented diagrammatically as having all its peaks and troughs as plane, parallel lines perpendicular to the direction of travel. The progression from spherical waves to plane waves can be seen in the sketch below. Close to the source, light has a spherical character. Far from the source, light has a planar character.



LAB #2: Diffraction

It will be stated without proof at this time that the angular distance, θ_1 , the distance from the center of the narrow slit of width d , to the position of the first dark minima is given by:

$$\sin \theta_1 = \frac{\lambda}{d}$$

Distance from central maximum to first minimum

where λ is the wavelength of the incident light. Since the distance D is much larger than the slit dimension d , we can replace the sine function with the numerical value of the angle, measured in radians ($1 \text{ rad} = \pi/180^\circ$). For this special case,

$$\theta_1 = \frac{\lambda}{d} \text{ radians}$$

Angular Distance from central maximum to first minimum

Again, since in our study of diffraction, $D \gg d$, (a "thin" slit) the linear separation between dark diffraction minima for a single slit can be closely approximated as:

$$x = D \left(\frac{\lambda}{d} \right)$$

Linear Separation between Adjacent Minima

and the linear distance between the center of the aperture to any minima is given by:

$$x' = m \cdot D \left(\frac{\lambda}{d} \right)$$

Distance from Center of Central Maximum (center of slit) to any Minimum

where m is an integer > 0 ($m = 1, 2, 3, \dots$). The value $m = 0$ is the center of the central maximum.

Diffraction by a Circular Aperture:

From geometry, we know that a circle is a collection of all points in a plane at a fixed distance from a fixed point on that plane. Imagine that a circular aperture is a slit of separation d , rotated about a line through the center of the slit and in the direction of the light. The aperture will then have a diameter of d .

From this simple argument, we would expect that the diffraction pattern of a circular aperture would be very much the same as from a single slit. Indeed, a central bright circular spot is observed on a screen surrounded by concentric dark minima. The radii of the successive dark minima are:

Central Maximum	$\theta_0 = 0 \cdot D \cdot (\lambda/d)$
First Dark Radius	$\theta_1 = 1.220 \cdot D \cdot (\lambda/d)$
Second Dark Radius	$\theta_2 = 2.233 \cdot D \cdot (\lambda/d)$
Third Dark Radius	$\theta_3 = 3.238 \cdot D \cdot (\lambda/d)$
Fourth Dark Radius	$\theta_4 = 4.241 \cdot D \cdot (\lambda/d)$

4.) Procedure

Single Slit Diffraction:

1. Mount the air slit in front of the laser (6-10 in).

2. Attach three 8.5x11 papers to the wall opposite from the HeNe laser using masking tape.
3. Unblock the HeNe laser.
4. Be certain the laser beam passes through the air slit aperture and strikes the paper.
5. Adjust the position of the mirror mount until a single slit diffraction pattern is observed on the paper. Note: the room lights might require dimming to observe this phenomenon.
6. Measure the separation distance, D_{SLIT} , between the aperture slit and the screen and record it.
7. Carefully mark the positions on the central maxima and all the dark diffraction minima on the paper.
8. Measure the distance from the central maxima to succeeding diffraction minima and record these positions in Table 1.
9. Sketch the diffraction pattern for this slit separation.
10. Block the laser beam.
11. Calculate the positions of the dark diffraction minima using the measured distance between the aperture and viewing screen (paper). Record the results in Table 1 (HeNe = 632.8 nm).
12. Calculate the percent error between the calculated and experimentally measured dark diffraction minima for the air slit width.

$$\text{Percent Error} = \frac{|\text{measured} - \text{calculated}|}{\text{calculated}} \cdot 100\%$$

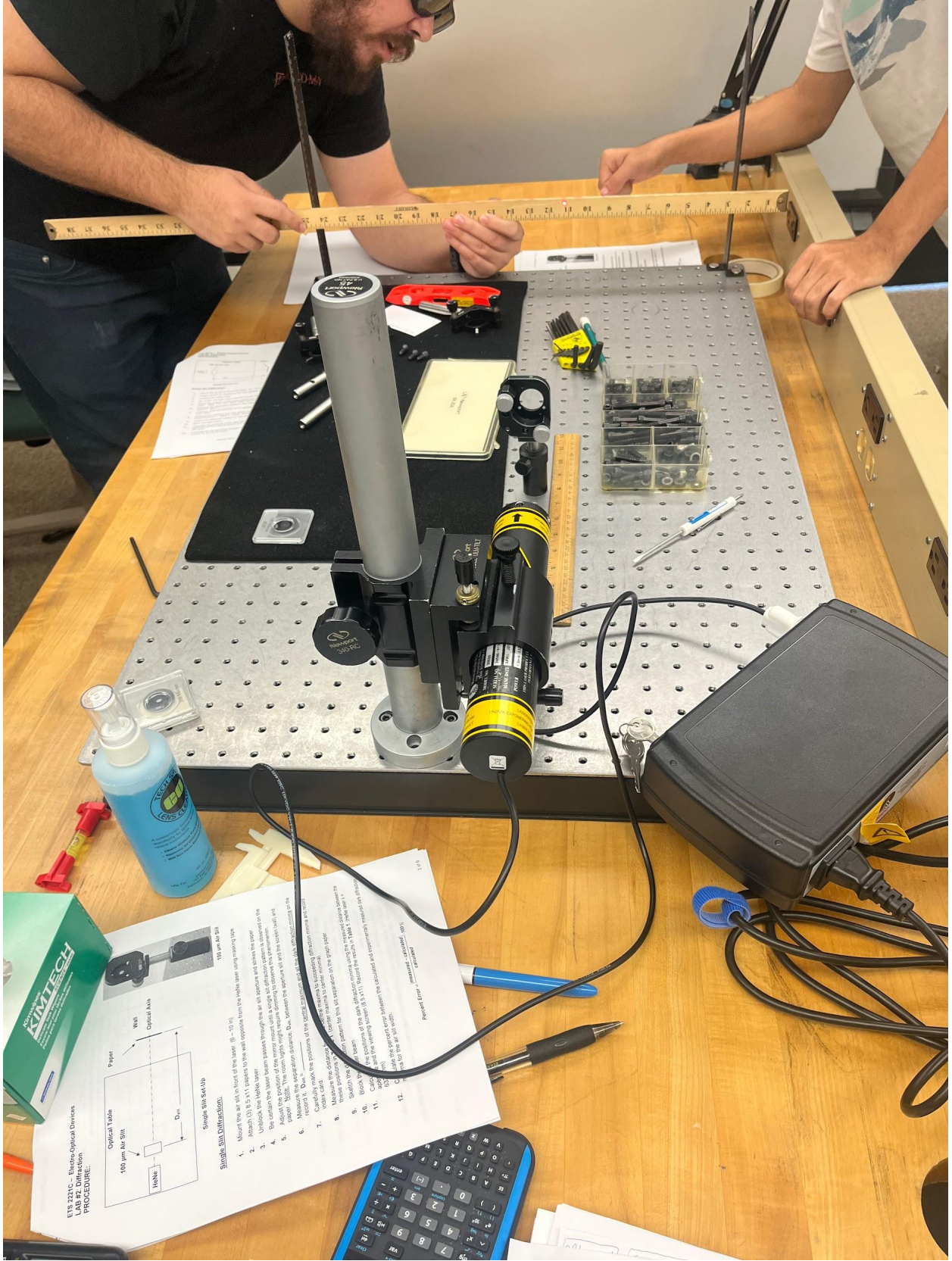
Circular Aperture Diffraction:

- 1.) Remove the air slit aperture.
- 2.) Mount the 300 l'm pinhole aperture.
- 3.) Attach (1) 8.5X11 paper to wall.
- 4.) Be certain the laser beam passes through the pinhole aperture and strikes paper.
- 5.) Adjust the paper until a diffraction pattern can be observed. Room light may require dimming to observe pattern.
- 6.) Measure separating distance between the aperture slit abs screen and record it.
- 7.) Carefully mark the positions on the central maxima and all the dark diffraction minima on the paper. Measure the distance from the central

maxima to succeeding diffraction minima and record these positions in Table 2.

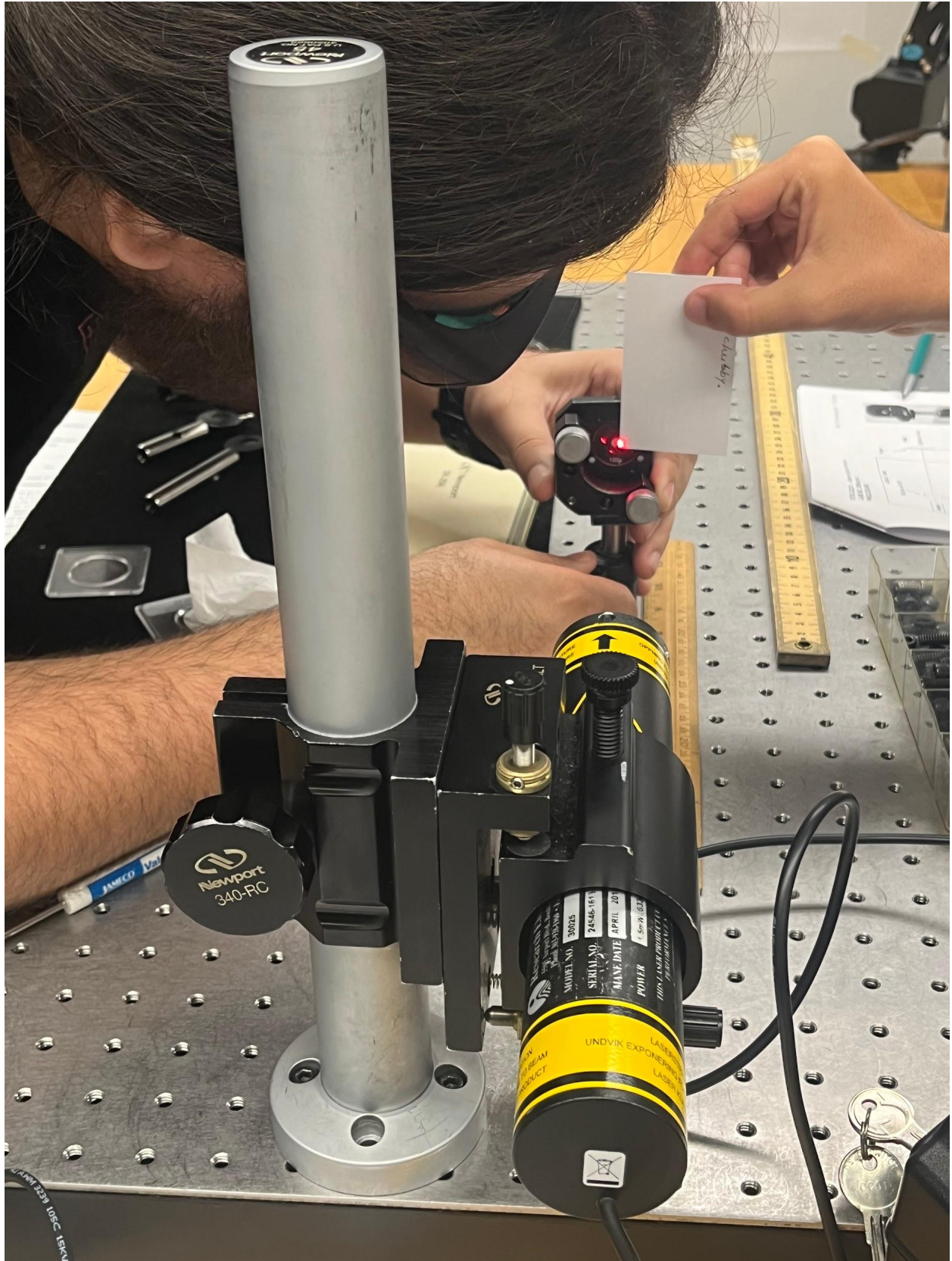
- 8.) Sketch the diffraction pattern for this slit separation.
- 9.) Block the laser beam.
- 10.) Calculate the positions of the dark diffraction minima using the measured distance between the aperture and viewing screen (paper). Record the results in Table 2 (HeNe = 632.8 nm).
- 11.) Calculate the percent error between the calculated and experimentally measured dark diffraction minima for the air slit width.

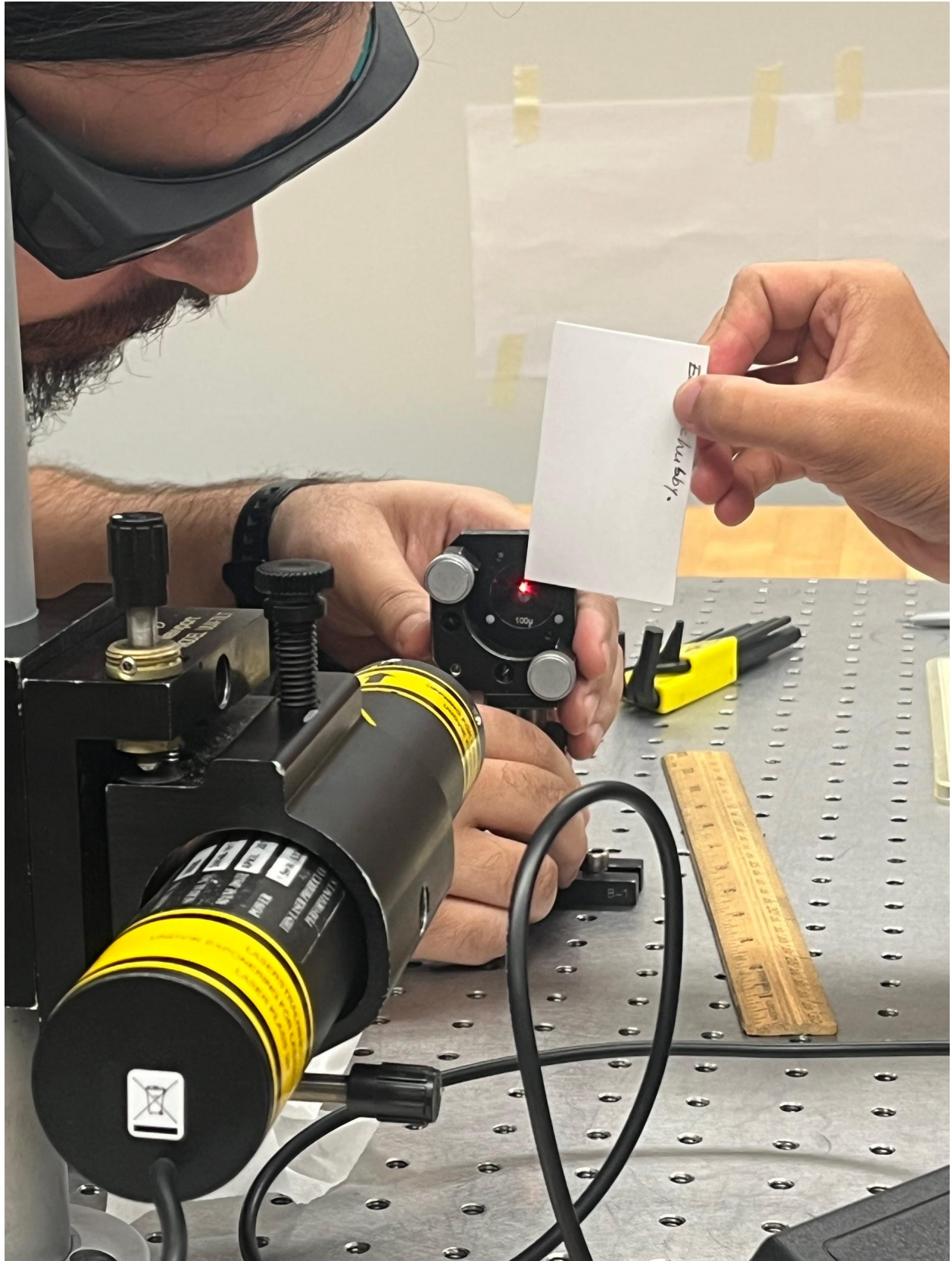
$$\textbf{Percent Error} = \frac{|\textbf{measured} - \textbf{calculated}|}{\textbf{calculated}} \cdot 100\%$$



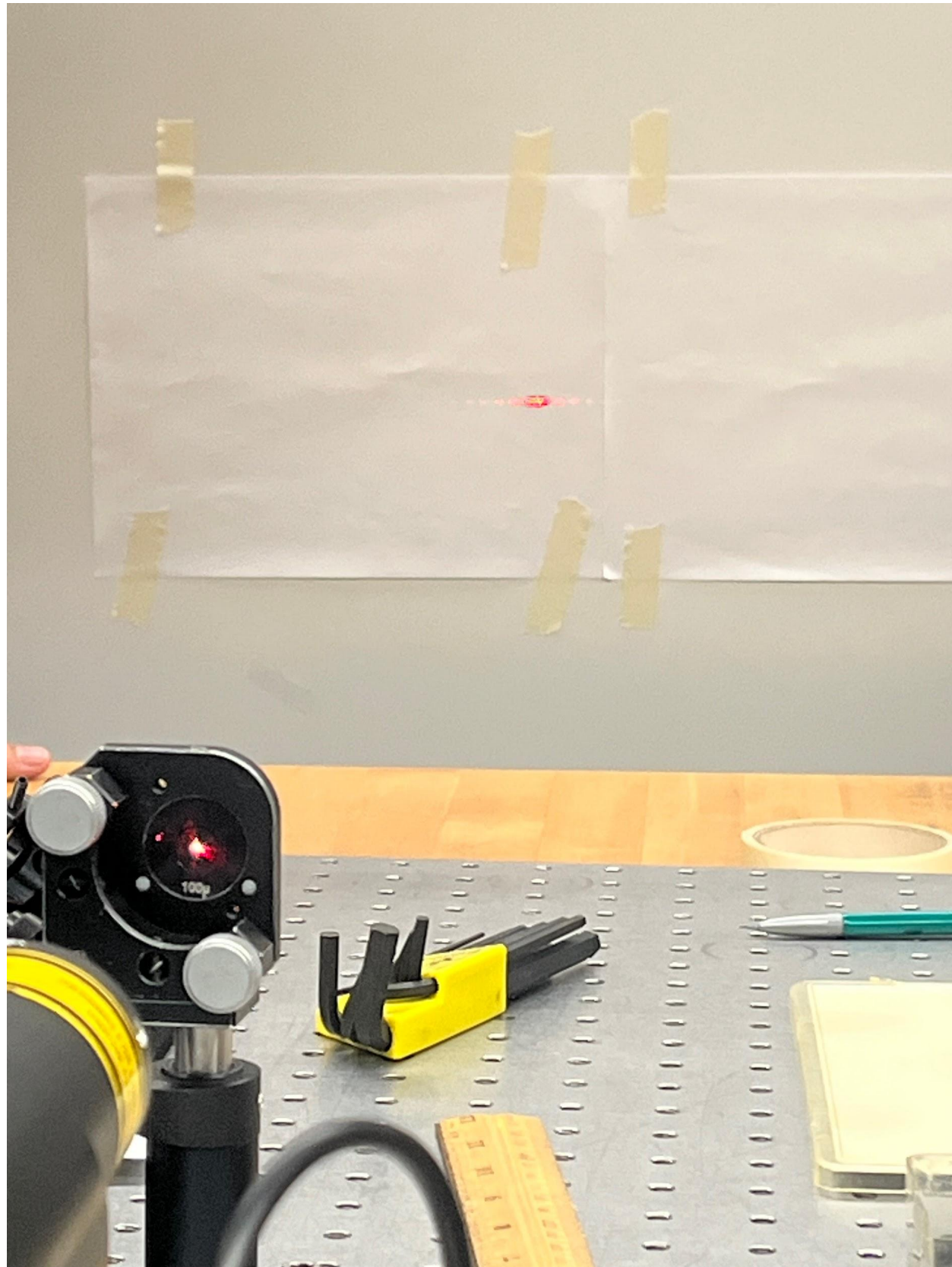


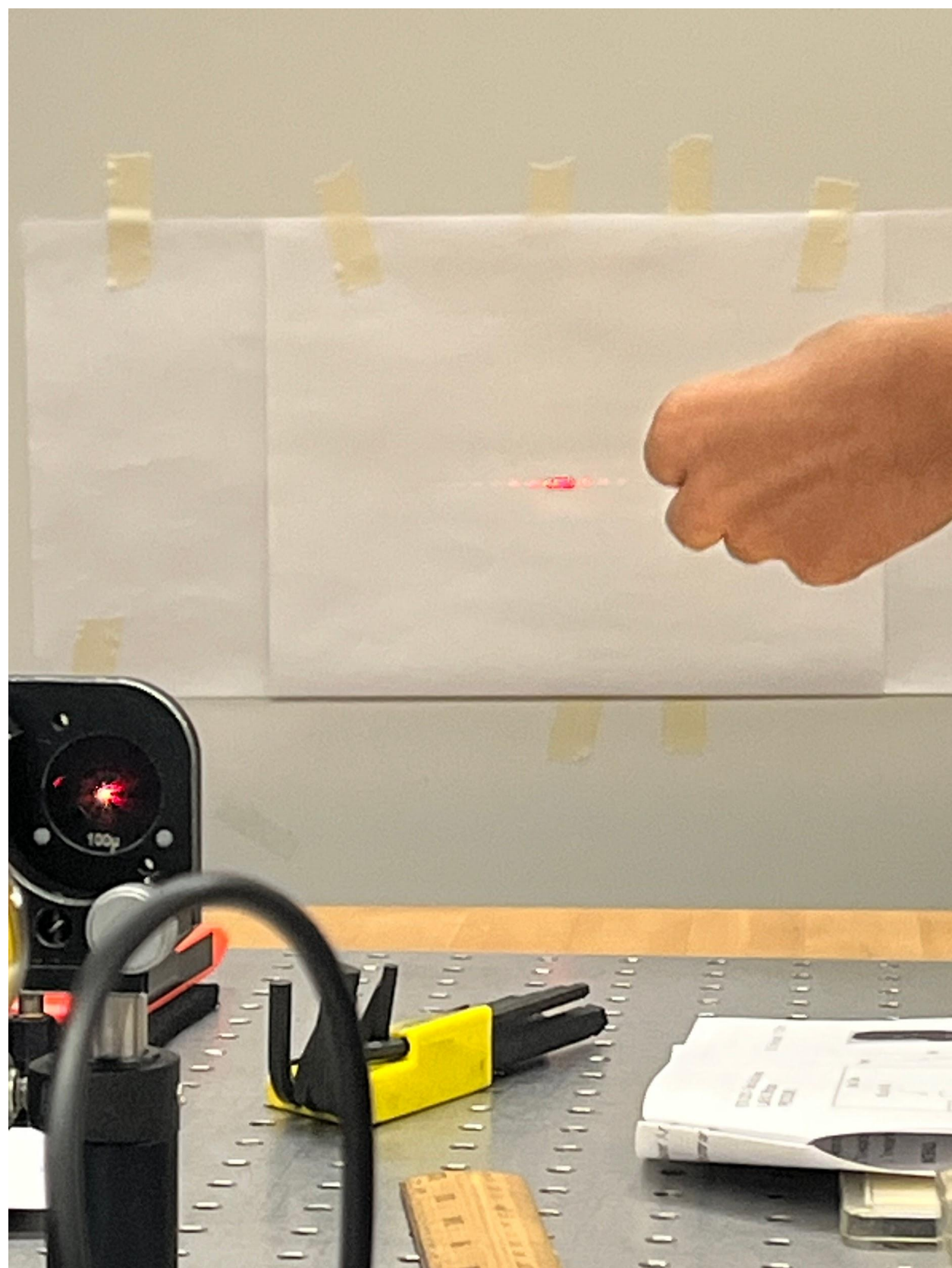
Lab setup for both pictures.

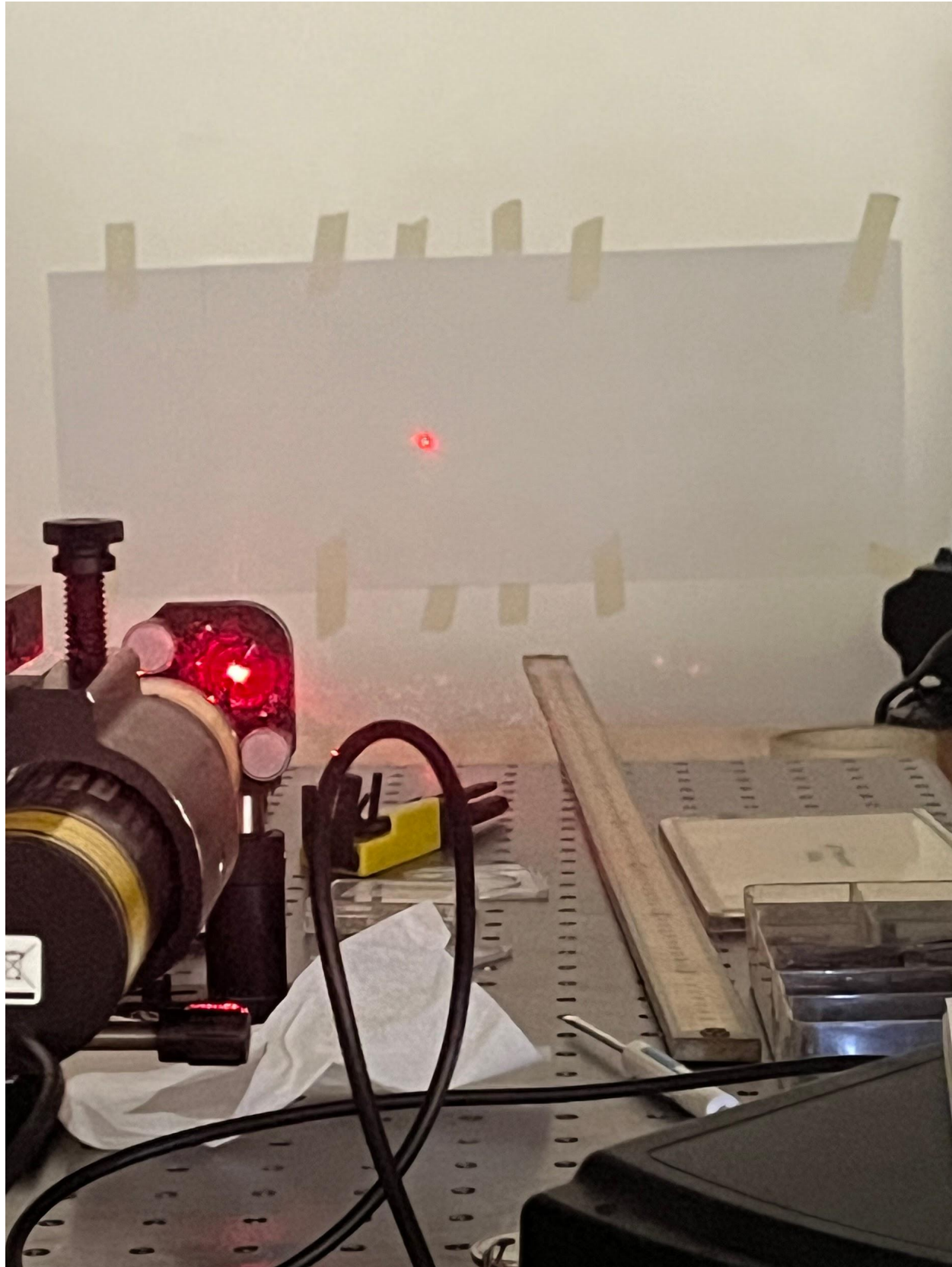




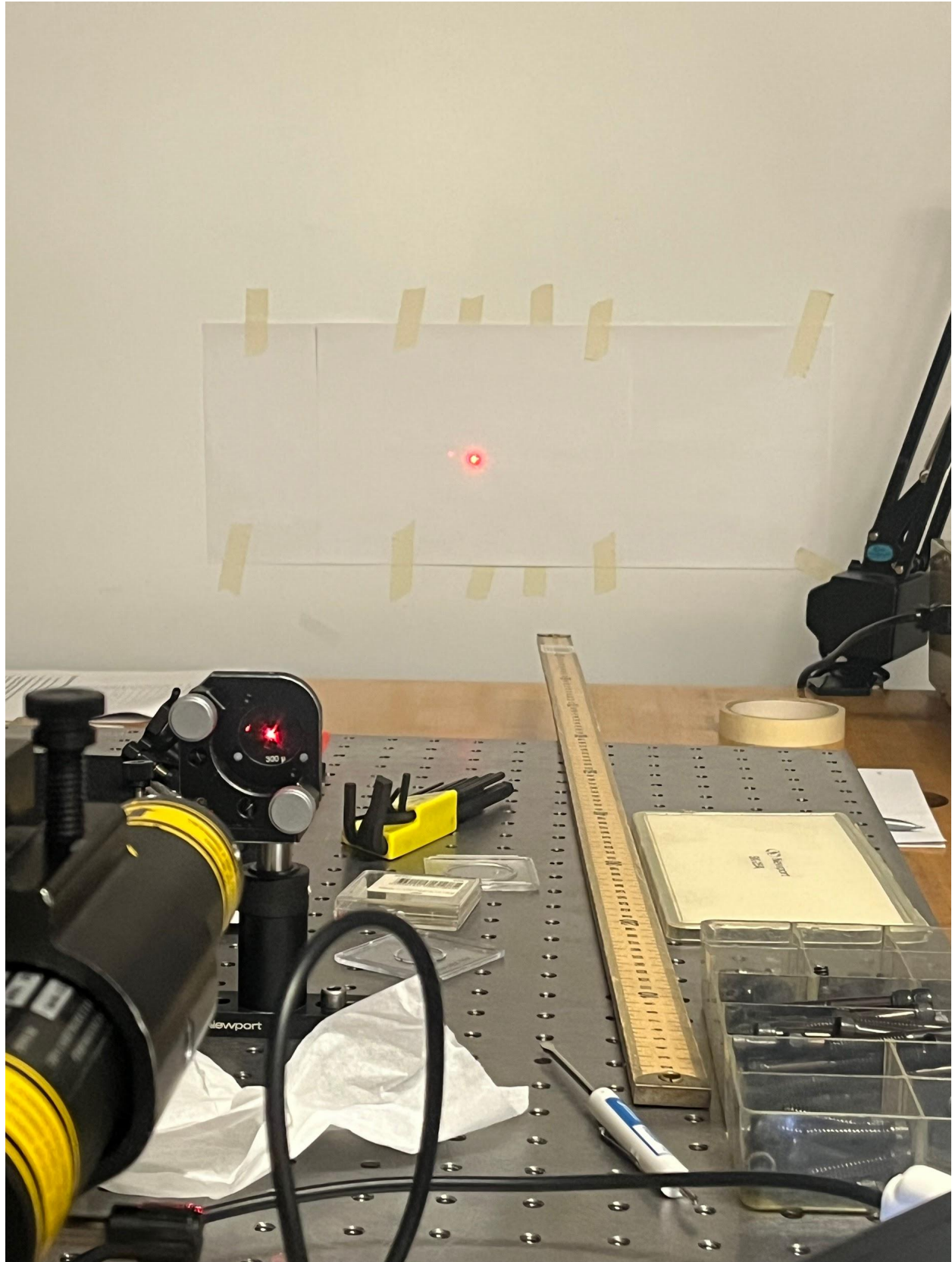
Measuring central maxima and dark diffraction.







Observing pattern on paper screen



Measuring distance.

5.) Data & Observations

TABLE 1 Single Slit

Position	m value	Measured cm	Calculated cm	% Error
Left	-5	4.2	4.1132	2.11%
Left	-4	3.3	3.2906	0.29%
Left	-3	2.5	2.4679	1.30%
Left	-2	1.6	1.6453	2.75%
Left	-1	0.8	0.8226	2.75%
Center	0	Max	Max	2.75%
Right	1	0.8	0.8226	2.75%
Right	2	1.6	1.6453	2.75%
Right	3	2.4	2.4679	2.75%
Right	4	3.2	3.2906	2.75%
Right	5	4.0	4.1132	2.75%

$4.2 - 4.1132 / 4.1132 (100) \rightarrow 2.11$
 $3.3 - 3.2906 / 3.2906 (100) \rightarrow 0.29$
 $2.5 - 2.4679 / 2.4679 (100) \rightarrow 1.30$
 $1.6 - 1.6453 / 1.6453 (100) \rightarrow 2.75$
 $0.8 - 0.8226 / 0.8226 (100) \rightarrow 2.75$
 $0.8 - 0.8226 / 0.8226 (100) \rightarrow 2.75$
 $1.6 - 1.6453 / 1.6453 (100) \rightarrow 2.75$
 $2.4 - 2.4679 / 2.4679 (100) \rightarrow 2.75$
 $3.2 - 3.2906 / 3.2906 (100) \rightarrow 2.75$
 $4.0 - 4.1132 / 4.1132 (100) \rightarrow 2.75$

TABLE 2 Circular Aperture

Position	m value	Measured cm	Calculated cm	% Error
Left	4.241	1.1	1.16	5.17%
Left	3.238	0.85	0.889	4.39%
Left	2.233	0.6	0.613	2.12%
Left	1.220	0.35	0.335	4.48%
Center	0	Max	Max	
Right	1.220	0.35	0.335	4.48%
Right	2.233	0.6	0.613	2.12%
Right	3.238	0.9	0.889	1.24%
Right	4.241	1.15	1.16	0.862%

$$1.1 - 1.16 / 1.16 (100) \rightarrow 5.17$$

$$0.85 - 0.889 / 0.889 (100) \rightarrow 4.39$$

$$0.6 - 0.613 / 0.613 (100) \rightarrow 2.12$$

$$0.35 - 0.335 / 0.335 (100) \rightarrow 4.48$$

$$0.35 - 0.335 / 0.335 (100) \rightarrow 4.48$$

$$0.6 - 0.613 / 0.613 (100) \rightarrow 2.12$$

$$0.9 - 0.889 / 0.889 (100) \rightarrow 1.24$$

$$1.15 - 1.16 / 1.16 (100) \rightarrow 0.862$$

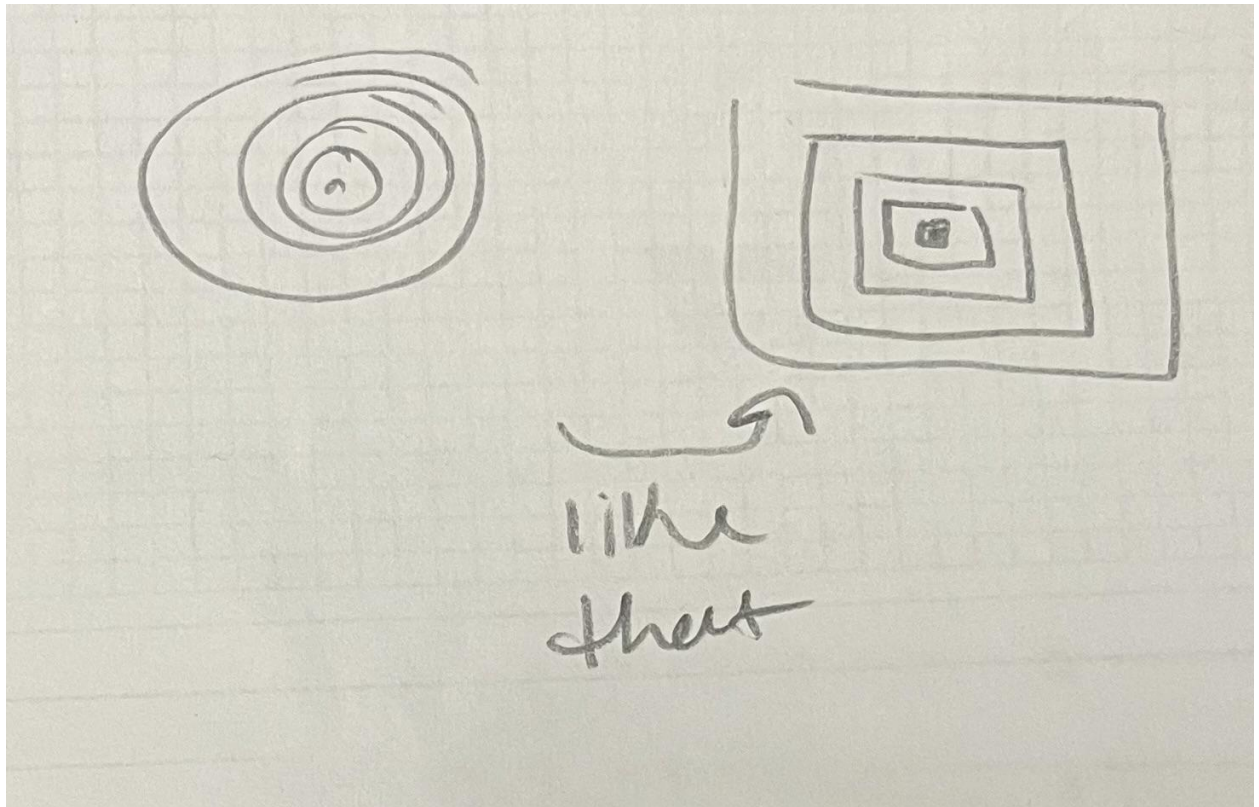
The data obtained is as accurate as possible by measurement tools and the conductors of the experiment. This data shows that the different diffraction patterns of the slit and the pinhole aperture are accurately observed. Some errors can occur during the experiment if the power, measured distance, or calculations are incorrect.

6.) Answers to Lab Questions

1.) How does the aperture size affect the pattern? Smaller aperture gives you a larger pattern.

2.) What would you expect the diffraction pattern to be off instead of either a single slit or circular aperture, a small square aperture were used

instead? It would be more blocky.



7.) Conclusion

In conclusion, the laboratory experiment showed that both aperture patterns and measurements were accurate and consistent. The experiment yielded expected results because the measured values and the calculated values were very close to each other, they weren't vastly different values. This shows that the conductors of the experiment were very accurate in their measurements. Another observation were the diffraction patterns themselves. We observed the diffraction of the single slit and the diffraction of the circular aperture and the patterns were consistent and what we were expecting. Improvements would be to ensure that user errors and equipment errors are at a minimum so that the results are as accurate as possible for future work.